3.1. UNIQUENESS OF DELEGATED PORTFOLIO MANAGEMENT AGENCY ISSUES

A large part of what portfolio managers do is acquire information. The manager who takes a fundamental approach pores over company reports and visits firms to gain insights into a company, while a quant manager writes code to analyze data to forecast future stock price movements. Thus portfolio management typically involves the search for an information signal, which can be the basis for action. This makes the agency problems in portfolio management different from standard agency problems, which are usually about direct performance—for example, how many widgets is the factory worker making? In standard agency problems, the agent does not acquire information and only decides how many widgets to make. The second difference from standard agency problems is that the fund manager controls both returns and risk of the portfolio. In the classical formulations of agency problems, the agent usually only controls a rate of output (how many widgets do you make on average?), not its variability (what's the range of widgets we're likely to make today?).[8]

What the asset owner can observe is also different from the standard principal-agent setup. In the standard moral hazard problem, *effort* or *action* is unobservable. The principal is unsure of how much the factory worker is slacking or working. But the factory owner does know that the worker is either slacking or working, or more generally somewhere in the spectrum between totally slacking off or all-out working. The *choice set* or the *action set* of the worker is limited. In the portfolio management problem, the asset owner does not know the full set of portfolios that could have been selected but (sometimes) observes the final portfolio selected. That is, the reverse of the standard principal-agent set-up occurs: *the asset owner observes the action chosen but not the action set*. The very first paper in the delegated portfolio management field, Bhattacharya and Pfleiderer (1985) solved these issues and found that the principal needed a *nonlinear* contract to hire the best agents and ensure the best agents made the right portfolio selection.

The marked differences between standard principal-agent problems and delegated portfolio management problems have caused the field of delegated portfolio management to develop relatively slowly compared to other areas of contract theory. In Stracca's (2006) excellent survey, he says that because of these differences, "the literature has reached more negative rather than constructive results and the search for an optimal contract has proved to be inconclusive even in the most simple settings."[9] Stracca is correct—economists have not yet developed a sure-fire

[8] In some problems the agent controls the variance (what's the maximum and minimum number of widgets you make each shift?), but the agent does not control both the mean and variance, which the fund manager does.

[9] See also the literature survey by Bhattacharya et al. (2008). A somewhat dated, but lucid, review is by Lakonishok et al. (1992).